

ALLEN COUNTY AP, OH, USA

WMO: 724298

Lat: 40.708N

Lon: 84.027W

Elev: 297

StdP: 97.81

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 04850

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-14.1	-21.2	0.6	-16.4	-18.3	0.8	-13.5	12.8	-3.9	11.9	-2.1	3.9	240	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	32.3	22.9	31.1	22.5	29.5	21.8	24.9	29.8	24.0	28.5	23.2	27.4	4.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.6	27.3	22.5	17.8	26.6	22.0	17.3	26.0	76.8	29.7	73.2	28.6	70.1	27.3	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.5	8.5	-20.2	34.2	3.4	2.2	-22.6	35.8	-24.6	37.1	-26.5	38.3	-29.0	39.9		
(5)			-20.6	26.2	3.3	1.2	-23.0	27.1	-24.9	27.7	-26.7	28.4	-29.1	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-2.9	-1.5	3.8	10.7	16.7	21.4	23.0	22.1	18.7	12.3	5.9	0.0
	DBStd	10.51	6.74	6.36	6.19	5.21	4.81	3.69	3.07	2.84	4.13	5.07	5.39	5.94
	HDD10.0	1487	401	324	211	52	5	0	0	0	1	33	147	314
	HDD18.3	3193	657	554	451	233	91	14	2	4	44	198	374	569
	CDD10.0	1823	2	4	19	74	212	342	404	374	262	104	23	3
	CDD18.3	486	0	0	1	6	39	107	147	119	56	11	0	0
	CDH23.3	4067	0	0	7	50	337	888	1298	919	493	75	1	0
	CDH26.7	1251	0	0	1	5	85	291	445	262	149	13	0	0
Wind	WSAvg	4.1	5.1	5.0	4.7	4.9	4.1	3.6	3.0	2.7	3.0	3.8	4.5	4.8
Precipitation	PrecAvg	971	65	51	69	93	99	106	103	88	85	67	68	72
	PrecMax	1362	184	145	148	230	170	328	266	232	167	134	138	191
	PrecMin	706	19	14	24	32	19	33	37	24	31	20	8	24
	PrecStd	171	34	31	29	39	45	59	54	44	40	32	38	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.2	18.5	23.6	27.4	31.3	33.5	34.5	32.8	32.7	28.2	22.1	16.4
		MCWB	13.0	13.5	16.5	18.1	20.8	22.1	23.7	23.4	21.7	20.2	15.2	13.3
	2%	DB	12.4	14.1	19.5	24.7	28.9	31.7	32.4	31.1	30.4	25.3	18.4	13.6
		MCWB	10.9	11.8	14.3	16.4	19.8	22.1	23.6	23.1	21.8	17.8	14.1	11.5
	5%	DB	9.6	11.0	16.8	22.5	27.2	30.0	31.0	29.3	28.1	22.8	16.6	11.0
		MCWB	7.9	8.5	12.3	15.1	19.4	21.5	22.8	22.5	20.4	17.0	12.6	9.2
	10%	DB	6.3	7.5	13.7	19.9	25.0	28.0	29.1	27.8	26.3	20.8	14.1	8.0
		MCWB	4.2	5.2	10.0	13.7	18.5	20.6	22.1	21.8	19.2	16.2	10.8	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	15.0	17.0	19.9	22.6	24.8	26.1	25.7	24.1	21.8	17.2	14.3
		MCDB	14.8	17.6	23.1	25.2	28.1	30.2	31.2	30.2	29.4	27.2	19.4	15.8
	2%	WB	11.2	11.9	14.9	17.8	21.4	23.8	25.1	24.6	22.9	19.9	15.0	11.4
		MCDB	12.4	13.6	18.8	22.7	26.8	28.9	30.4	29.0	27.9	23.5	17.7	13.1
	5%	WB	7.9	8.5	13.0	16.1	20.3	22.9	24.2	23.7	21.8	18.3	13.2	9.1
		MCDB	9.3	10.3	16.3	20.7	25.3	27.9	28.6	27.7	26.2	21.5	15.8	10.8
	10%	WB	4.2	5.2	10.4	14.8	19.3	21.9	23.3	22.8	20.7	16.5	11.3	6.5
		MCDB	6.0	7.4	13.4	19.3	23.8	26.5	27.4	26.4	24.5	20.3	14.1	8.1
Mean Daily Temperature Range	5% DB	MDBR	7.6	7.9	9.2	11.2	10.9	10.4	10.5	10.6	11.6	10.3	8.7	6.8
		MCDBR	9.7	11.4	12.7	13.9	12.7	12.1	11.4	11.8	13.4	12.7	11.3	9.5
		MCWBR	8.6	9.5	8.5	8.0	6.4	5.3	4.8	5.4	6.2	7.0	8.0	8.6
	5% WB	MCDBR	9.5	10.9	11.4	12.2	10.8	10.0	9.8	9.7	10.7	10.4	10.3	9.3
		MCWBR	9.0	9.6	8.1	8.1	6.1	5.2	4.9	5.3	5.6	6.6	8.3	8.9
Clear-Sky Solar Irradiance	taub	0.307	0.325	0.347	0.399	0.440	0.444	0.445	0.442	0.398	0.357	0.324	0.301	
	taud	2.389	2.333	2.353	2.218	2.148	2.187	2.202	2.194	2.289	2.393	2.442	2.443	
	Ebn at Noon	848	883	903	873	842	836	831	823	836	837	822	828	
	Edh at Noon	83	101	112	137	150	145	142	139	118	95	79	73	
All-Sky Solar Radiation	RadAvg	1.60	2.54	3.51	4.68	5.47	6.05	6.04	5.42	4.40	2.99	1.95	1.35	
	RadStd	0.15	0.30	0.31	0.41	0.46	0.45	0.34	0.24	0.36	0.26	0.26	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+101	N/A

Nomenclature: See separate page